Implementation of a Biomarker-Preserving Bone Biopsy Decalcification Workflow for Oncology Patients: A Single-Center Quality Improvement Initiative

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Background

Accurate histopathologic classification and biomarker assessment from bone metastasis biopsies are critical for guiding systemic therapy across solid tumors, including breast, prostate, lung, and urothelial cancers. Immunohistochemistry (IHC) and related biomarker analyses frequently determine treatment selection. However, osseous specimens typically require decalcification prior to evaluation, and preanalytic processing can significantly impact antigen preservation. Acid-based decalcification, commonly used in routine practice, may degrade antigenicity and increase the risk of false-negative or nondiagnostic biomarker results. In contrast, EDTA-based decalcification preserves receptor integrity but is not consistently implemented. Despite its clinical importance, there is often no standardized workflow to identify cases requiring biomarker-preserving processing. This gap represents a potentially modifiable source of diagnostic inaccuracy and suboptimal oncologic care.

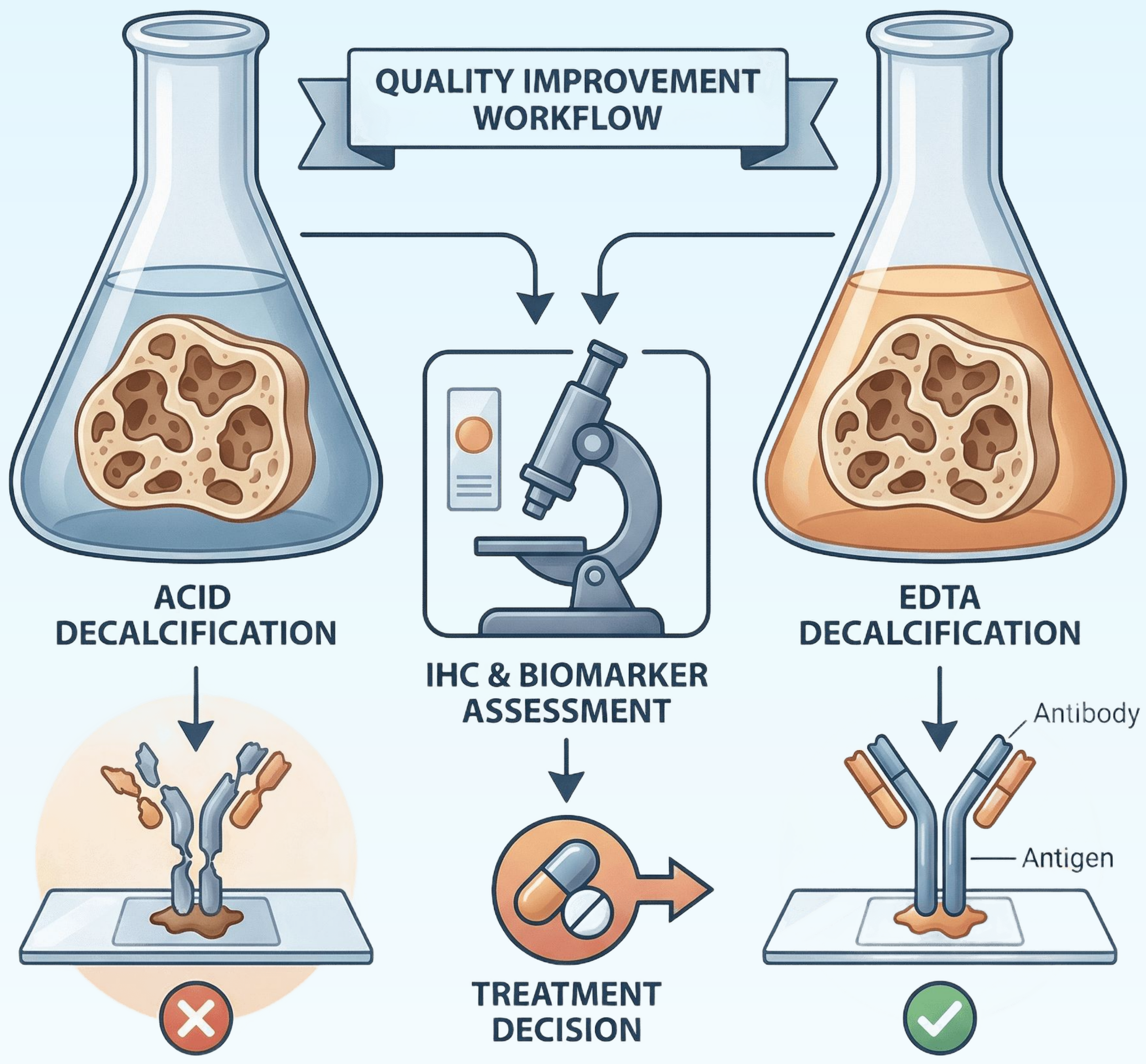
Objectives

The primary objective is to design and implement a standardized, biomarker-preserving decalcification workflow for patients undergoing image-guided bone biopsy for suspected metastatic malignancy. Secondary objectives include:

- Evaluating baseline institutional decalcification practices and their impact on biomarker adequacy
- Identifying rates of nondiagnostic or discordant IHC results
- Assessing downstream clinical consequences, including repeat biopsies and treatment modifications

Proposed Methods

This is a single-center, multidisciplinary quality improvement initiative. A retrospective baseline analysis will include patients undergoing image-guided bone biopsy for suspected metastatic solid tumors from January 2023 to December 2025. Cases requiring treatment-informing IHC or lineage-defining biomarkers will be evaluated for decalcification method, adequacy of biomarker testing, concordance with clinical and pathologic expectations, and need for repeat procedures or external review.



Anticipated Contributions

This study aims to establish a scalable, biomarker-preserving workflow for bone biopsy processing that improves diagnostic accuracy and enhances treatment selection. By addressing a key preanalytic variable, this initiative has the potential to reduce repeat procedures, improve patient outcomes, and standardize best practices across institutions.

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